

Product Requirements Document

Property Value & Development Impact Estimator — Telangana

Status: Draft v1 | Type: Capstone Project

1. Problem Statement

People buying, selling, or holding land/homes/flats in Telangana routinely lack visibility into upcoming government development plans (metro extensions, highways, IT corridors, SEZs, industrial parks) that materially affect property value. Without this, buyers overpay for areas about to stagnate and sellers underprice land about to appreciate. Existing real estate portals show only current listing prices, with no forward-looking, news-grounded context.

2. Goals

- Give any Telangana resident a map-based view of approximate current property value by region.
- Surface a transparent, explainable projection of how that value may shift, driven by real government/infrastructure development news — not a black-box prediction.
- Always show the source articles behind a projection so users can verify the reasoning themselves.

3. Non-Goals

- This is **not** a certified valuation tool, investment advisory product, or replacement for professional appraisal/legal due diligence.
- No transaction facilitation (no buying/selling/brokerage features) in v1.
- No user-submitted listings or crowdsourced pricing in v1.
- No property-level (single house/flat) valuation — v1 operates at region (mandal/zone) granularity.

4. Target Users

- Individual buyers/sellers of land, homes, or flats in Telangana without insider market knowledge.
- Small investors comparing regions before committing capital.
- (Secondary) students/researchers interested in development-driven price trends.

5. Scope

Geography: Whole Telangana state (not limited to Hyderabad), so regional contrast in development activity is visible across the state, not just within one city.

Seed data: ~20-50 regions curated at launch as a cross-state sample — mixing high-activity Hyderabad-periphery zones (ORR/RRR corridor) with a couple of Tier-2 city mandals (e.g.

Warangal, Karimnagar) so the demo shows genuine statewide contrast. Full mandal-level coverage (600+ mandals) is future work, not v1.

6. Core Features

6.1 Interactive map

- Region boundaries (mandal/zone) rendered on a map, color-coded by projected value change (up / flat / down).
- Click a region to open a detail panel.

6.2 Region detail panel

- Approximate current value (per sq yd / sq ft), sourced from government market value (circle rate) data.
- Projected value shown as a **range**, not a single number, with a confidence indicator.
- Explanation trail: list of the specific news articles that drove the projection, each linked to its source.

6.3 Daily development-news pipeline

- Automated daily job fetches development-related news (metro, highway, IT park, SEZ, industrial corridor, etc.) relevant to Telangana.
- Each article is analyzed to identify: region(s) mentioned, project type, estimated timeline, and impact magnitude.
- An impact-scoring engine recalculates the projected value range for affected regions.

6.4 Trust & safety

- Persistent, non-dismissible-on-first-view disclaimer: *"Estimate based on public news, not financial advice."*
- Every projection is traceable to its contributing articles — no unexplained numbers.

7. Success Criteria (capstone demo)

- Map renders all seed regions with current values.
- At least one full day's automated news cycle runs end-to-end (fetch → extract → score → update) without manual intervention.
- At least 3 regions show a visibly different projection driven by distinct real news items, demonstrating the state-wide contrast goal.
- A user (evaluator) can click any projected value and see the exact articles behind it.

8. Risks & Mitigations

Risk	Mitigation
Users mistake projections for financial advice	Persistent disclaimer + explanation trail on every number
News-to-region matching is wrong (wrong mandal tagged)	Gazetteer + fuzzy matching layer; human-reviewable confidence score
Sparse news coverage for smaller mandals	Seed set weighted toward areas with genuine news activity; transparently show "no recent development news" rather than fabricating a projection
LLM extraction hallucinates project details	Structured JSON output only, always paired with the source link so a human can verify

9. Open Questions for Team

- Do we want a "watchlist" feature (user follows specific regions) in v1 or defer to v2?
- How often should base government market values be refreshed (they update periodically, not daily)?
- Do we log/store user interactions at all, or keep the app fully anonymous for v1?